Eco eco Rover

Electrical schematic and build guide

For	Electrical team, Team Eco eco, URRT Robo Cup
Revision	B, 6 Nov 2026
Repository	github.com/Jannatul-Naim/Eco_Eco
Rulebook	Mechanical & Electrical Rulebook V1.3
Pins come from	src/s3/config_pins.h, src/cam/config_cam.h, src/audio/config_audio.h
Rebuild this PDF	python3 tools/make_electrical_pdf.py

Status of this design

Build from this document, but know what it is. The power side (battery, fuses, contactor, E-stop, buck converters, motor drivers, motors) has **not been built or tested**. Section 10 lists exactly what has run on real hardware.

Before you buy or solder

- **Read the motor label.** Fuses, driver and wire sizes assume Pololu #4744 (5.5 A stall). The team's notes list 25GA 300 rpm with an unknown rating.
- **Confirm the driver.** The 10 A dual H-bridge in the notes is too small for the baseline motors. Use MD30C or BTS7960.
- **Never connect the 5 V sonar Echo directly to an S3 pin.** Use the divider on Sheet 5.
- **Never power the servo** from a board's 5 V pin or from the logic buck.
- **E-stop: the team chose a 2-pole switch** instead of the red mushroom-head button rule 1.2.1 asks for. Get the organisers' approval in writing before you build it. An inspector may refuse it, and a rover that fails the E-stop test may not run (rule 1.2.3).
- **No breadboard anywhere on the rover** (rule 3.1.2). The rover is not allowed to run with one.
- **Check your board's silkscreen** against Sheet 1. If a label differs, stop and ask before you solder.

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1. How to read this document

Label	Meaning
CONFIRMED	The team has said it owns this or has decided on it.
RECOMMENDED	Proposed here, with the reason given. Replace it if you have a better verified part.
UNKNOWN	Needed but not identified yet. Do not wire anything that depends on it until it is checked.

Reference letters

Ref	What	Ref	What
BT1	battery	U1	ESP32-S3 main controller (only motor authority)
F1, F3–F8	fuses (F2 not used)	U2	ESP32-CAM: video, QR, motion, servo
J1–J6	XT power connectors	U3	ESP32 DevKit: audio, optional
K1	contactor	U4, U5	motor drivers, left and right
SW1	E-stop switch	U6, U7	5 V buck converters, logic and servo
SW2	ARM button	U8, U9, LS1	microphone, amplifier, speaker, optional
D1	coil flyback diode	US1–US4	sonars (US, so they are not confused with the S3)
M1–M4	motors	SV1	MG996R pan servo
R, C	resistors, capacitors	N3	junction of ARM, hold path and E-stop pole A

How the parts fit together

- **S3 (U1)** is the only board that can command the motors. It runs missions, the sonars and the dashboard.
- **ESP32-CAM (U2)** handles video, QR, motion and the pan servo. It talks to the S3 over three wires, not Wi-Fi: S3 TX, S3 RX and GND.
- **DevKit (U3)** handles audio and talks over Wi-Fi only. It is optional.
- **The E-stop works without any of them.** SW1 cuts the K1 coil directly, so motors and servo lose power even if every board has crashed.

2. Decisions needed before buying

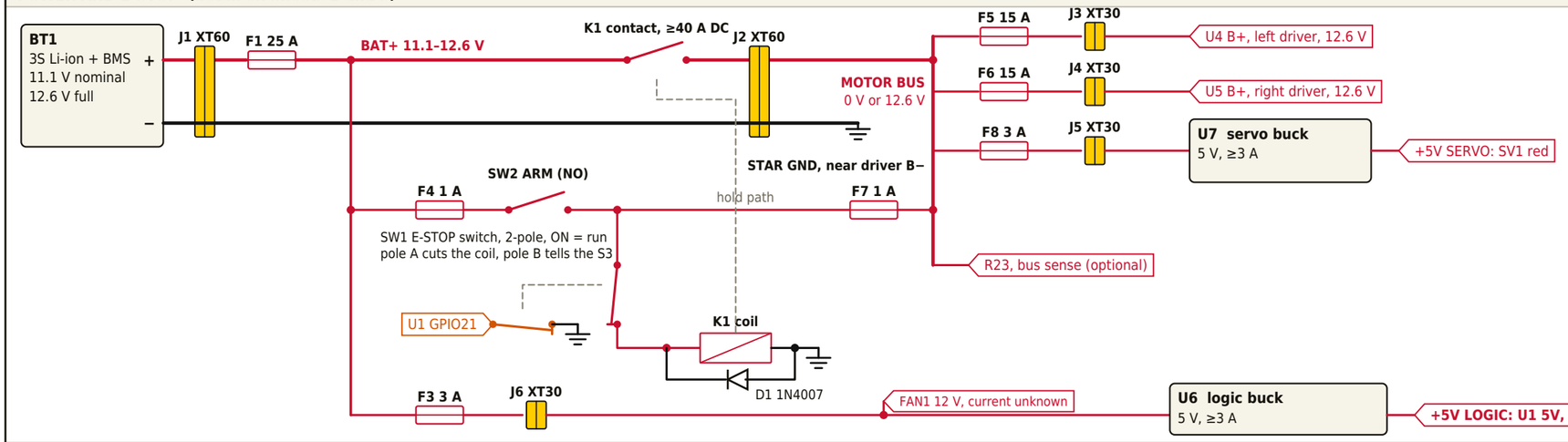
Item	Assumed here	Why it matters	How to settle it	Decided
Motors	Pololu #4744, 5.5 A stall	Sets every fuse, driver and wire size	Read the label or datasheet of the motors you will really fit: voltage, no-load and stall current	
Motor drivers	2 × MD30C	10 A dual H-bridge is below the 11 A per-side stall	Name the part you own. For BTS7960, change MOTOR_DRIVER_TYPE	
Battery	3S Li-ion, BMS ≥25 A	Stall worst case is 22 A	Choose and buy with its charger and bag	
Contactor K1	12 V coil, ≥40 A DC	Coil current sets F4, F7 and the E-stop contact rating	Read its datasheet. Prefer one with an auxiliary NO contact (Sheet 3)	
Fan	12 V 4-inch, on F3	Its current adds to the 3 A branch	Read the label	
Sonar count	4 planned, 1 owned	Reverse is blocked until the rear sonar is fitted	Buy at least the rear one next	
E-stop SW1	2-pole ON-OFF switch	Rule 1.2.1 asks for a red mushroom-head button; a rover that fails the E-stop test may not run	Ask the organisers in writing	Switch, team decision 6 Nov 2026

3. Rulebook checklist

Rule	Requires	This design	Where
1.2.1	Visible red mushroom E-stop, reachable from outside without crossing moving parts	**NOT MET** by team decision: SW1 is a 2-pole switch. Get the organisers' approval in writing	Sheet 3
1.2.2	Removes power from all actuators, in hardware, without software or wireless	SW1 cuts the K1 coil. Motors and servo are on the bus	Sheet 3
1.2.3	Tested at inspection	Tests T9 and T18	Section 9
1.2.4	Release must not restart motion	K1 latch needs ARM. Check the re-latch risk	Sheet 3, T10
3.1.1–3	Soldered veroboard or PCB, no breadboards, modules in soldered female headers, other joints soldered or crimped	BOM and build list	Section 7, 8
3.1.4	Insulated, tied down, strain relief, no exposed conductors	Build list	Section 8
3.1.5	Suitable wire gauge, motor wires apart from signals	Wire schedule gives each size	Section 5
3.2.1–2	XT or Deans on every power link, polarised battery connector	J1–J6	Sheet 2
3.2.3	Red positive with voltage labelled at each junction, black GND, other colours for signals	Colours in the wire schedule	Sheet 2, Section 5
3.3.1–3	≤24 V nominal, fused, secured, fire-resistant bag, no damaged packs	3S, F1, bag in the BOM	Sheet 2
3.4.2	Schematic shows controller, drivers, sensors, motors, battery, fuse, E-stop, every pin and rail voltage	Sheets 1–7, pin maps	Section 4, 6
3.4.3	Schematic and pin map in the GitHub repository	This PDF, in docs/	Repository
3.4.4	Built circuit must match the schematic before the M4 freeze	Change the sheets when the build changes	tools/ script

Rule 3.4.1 names schematic software or a neat pen-and-paper diagram. These sheets are drawn by a script. If a judge asks for a KiCad or EasyEDA project, redraw from Sheet 1.

POWER AND E-STOP (detail on Sheets 2 and 3)



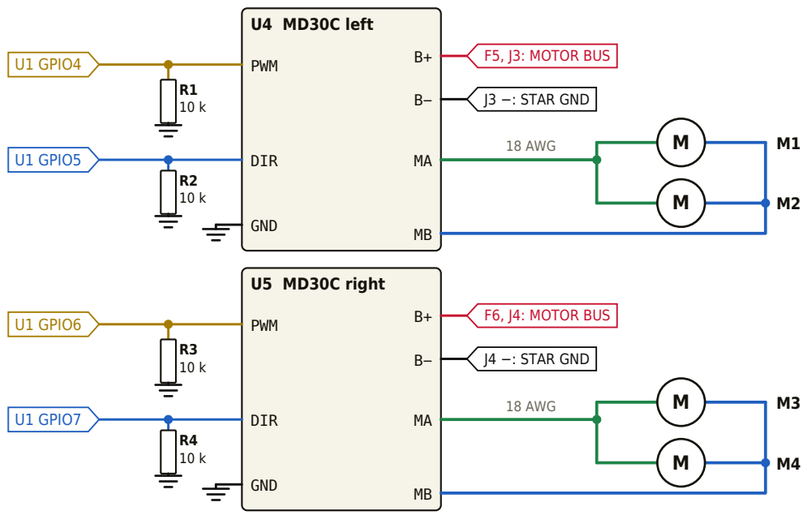
How to read this sheet

- Red: every positive line. Label the voltage at each junction.
- Black: GND and battery negative.
- Yellow: PWM. Blue: DIR.
- Green: sonar Trig. Violet: sonar Echo.
- Teal: camera link and voltage sense.
- Orange: servo signal and E-stop sense.

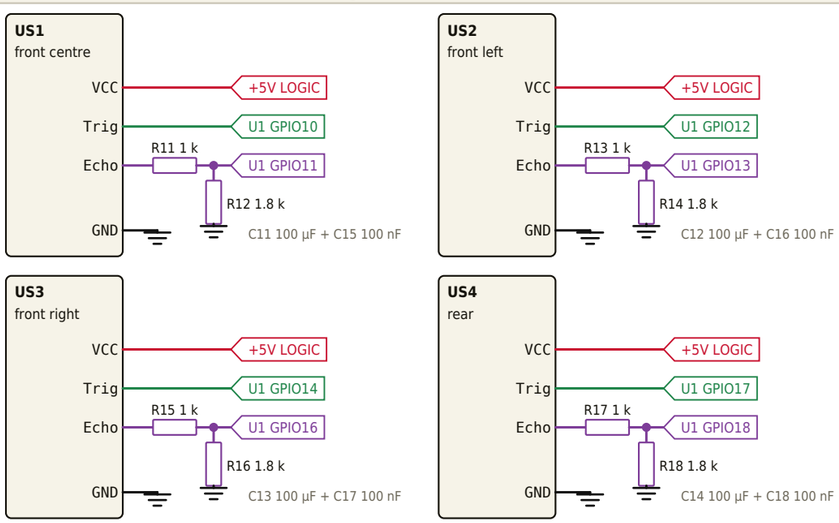
A tag points to the same net elsewhere on the sheet: follow the name. Crossing wires without a dot are not connected. Dashed grey is a mechanical link. Gold blocks are XT connectors.
Header order follows ESP32-S3-DevKitC-1. If your board's silkscreen differs from this sheet, stop and check before you solder.

Print on A3, or zoom in on screen.

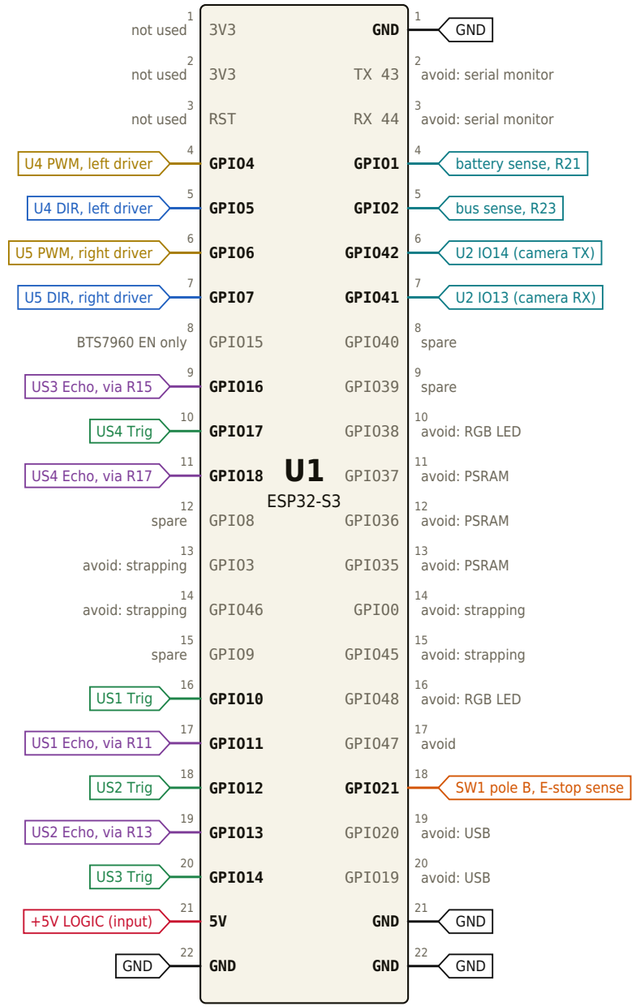
DRIVE: one MD30C per side, two motors in parallel (Sheet 4)



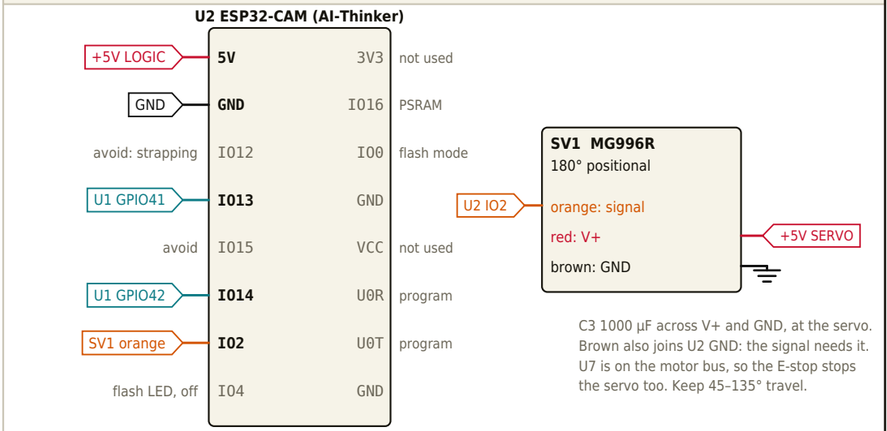
SONARS: HC-SR04 ×4, each Echo through its own divider (Sheet 5)



U1 ESP32-S3 MAIN CONTROLLER (N16R8, DevKitC-1 header)



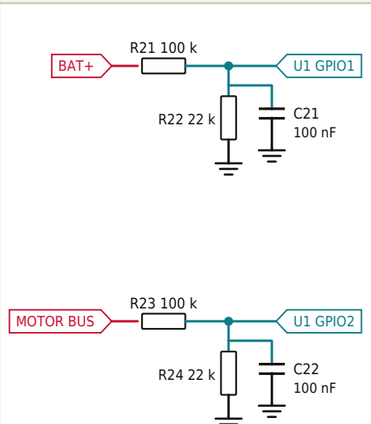
U2 ESP32-CAM AND SV1 MG996R PAN SERVO (Sheet 6)



C3 1000 µF across V+ and GND, at the servo. Brown also joins U2 GND: the signal needs it. U7 is on the motor bus, so the E-stop stops the servo too. Keep 45–135° travel.

C1 470 µF + C2 100 nF across 5V and GND, at the pin, short wires. Leave the microSD slot empty: it shares IO13 and IO14 with the link.

VOLTAGE SENSE, optional (Sheet 7)



U3 AUDIO BOARD, optional (Sheet 7)

U8 INMP441 mic
VDD: U3 3V3
GND, L/R: GND
SCK: GPIO26
WS: GPIO25
SD: GPIO34

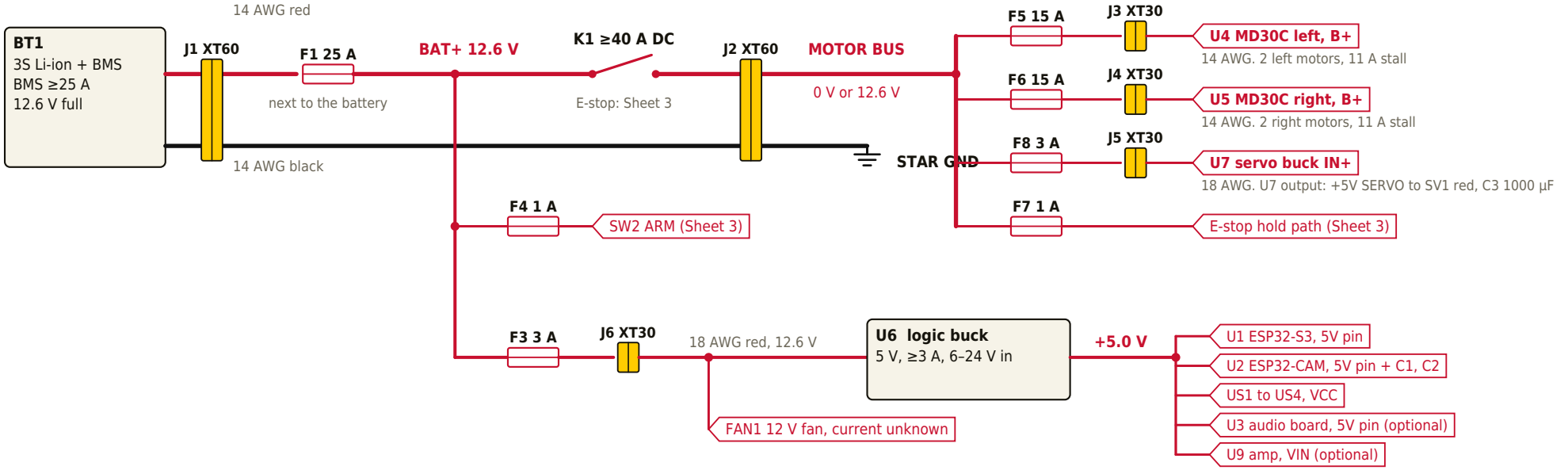
U9 MAX98357A amp
VIN: +5V LOGIC
BCLK: GPIO27
LRC: GPIO32
DIN: GPIO33
SD: GPIO23

U3 = ESP32 DevKit. 5V pin on +5V LOGIC, GND on GND. Its pin names above are its own GPIOs, not the S3's.

LS1 4 Ω 3 W speaker on U9 + and –.
U9 GND on GND, GAIN pin open (9 dB).

No wire to U1 except power and ground. It talks over Wi-Fi only.

Each junction shows its voltage, as rule 3.2.3 asks. Write the same label on the wire or the board.



STAR GND: run every return to one point near the driver B− terminals

U4 B−	U5 B−	U7 IN−	U6 IN−	K1 coil −, D1	FAN1 −	Logic board GND
J3 −, 14 AWG	J4 −, 14 AWG	J5 −, 18 AWG	J6 −, 18 AWG	22 AWG	22 AWG	22 AWG, one wire

Fuse	Protects	Rating	Wire	Why this size
F1	whole rover	25 A slow	14 AWG	22 A both sides stalled, plus logic and servo
F2	not used			number kept free to match docs/03_electrical.md
F3	logic buck + fan	3 A slow	18 AWG	about 1 A at 11.1 V, plus fan (check its label)
F4	ARM button path	1 A slow	22 AWG	K1 coil only. Check the coil current
F5, F6	each driver branch	15 A slow	14 AWG	2 × 5.5 A = 11 A per side at stall
F7	hold path	1 A slow	22 AWG	K1 coil only. Check the coil current
F8	servo buck	3 A slow	18 AWG	about 1.3 A at 11.1 V with the servo stalled

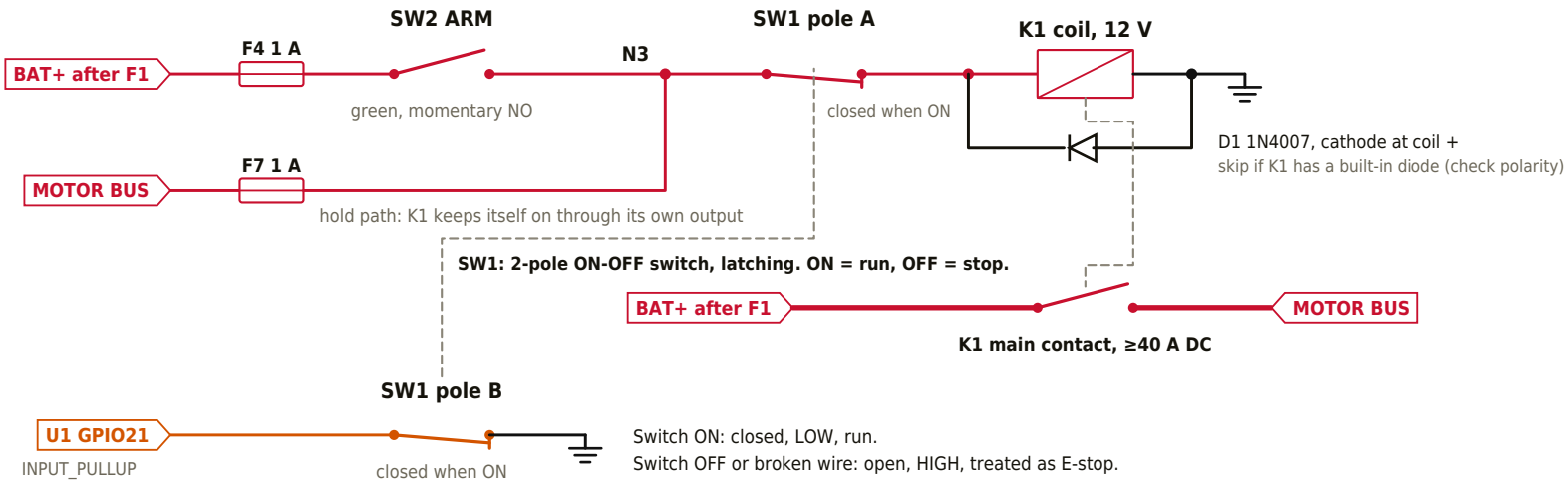
Before you buy

Fuses, driver and wire sizes assume Pololu #4744 (5.5 A stall). If you fit the 25GA motors from the team's notes, read their stall current first and redo this table. Slow-blow automotive blade fuses (ATO/ATC).

Never power the servo from a board's 5 V pin or from U6. Never use a generic 12 V adapter as the charger.

Connector	Between	Pin +	Pin −
J1 XT60	battery and F1	BAT+	GND
J2 XT60	K1 and motor bus	MOTOR BUS	GND
J3, J4 XT30	bus and U4, U5	after F5, F6	GND
J5, J6 XT30	U7, U6 inputs	after F8, F3	GND

J2 is added to what docs/03 first listed, so the E-stop assembly also meets rule 3.2.1: every power link between battery, E-stop, distribution and drivers uses an XT or Deans connector.



Action	What happens	Motor bus
Power on	K1 open. S3 boots with outputs disabled.	0 V
Press ARM (E-stop switch ON)	Coil on, K1 closes, then K1 holds itself from the bus.	12.6 V
Switch E-stop OFF	Pole A opens, coil drops, K1 opens. Pole B opens, S3 latches ESTOP.	0 V
Switch E-stop back ON	Nothing restarts. K1 stays open and S3 stays in ESTOP.	0 V
Restart on purpose	Switch E-stop ON, press ARM, then RESET E-STOP on the dashboard, then choose a mode.	12.6 V
S3 crash, Wi-Fi lost, dashboard closed	K1 is not affected. The E-stop still cuts power. S3 watchdog reboots with outputs LOW.	no change

Rules

Met, 1.2.2: SW1 cuts the coil directly, no software or wireless.

NOT MET, 1.2.1: the rulebook asks for a red mushroom-head button. The team chose a switch. Get the organisers' OK in writing before inspection.

Mount it outside, easy to reach without crossing moving parts. Label it E-STOP.

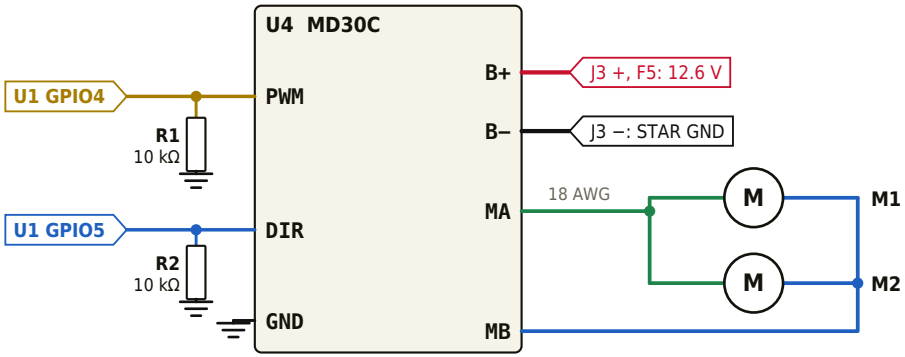
Design review: check this before inspection (not tested, no part fitted for it yet)

The hold path takes its power from the motor bus. After K1 opens, the bus can stay charged for a moment by the driver capacitors, and by the motors acting as generators while the wheels still turn. If someone switches the E-stop back ON in that moment, K1 could pull back in, which rule 1.2.4 forbids. The S3 would still hold outputs off, but the rule wants this in hardware. So: (1) run test T10 in Section 9 with the wheels spinning. (2) If K1 ever re-latches, fit one of these, and update this sheet. A contactor with an auxiliary NO contact is the clean fix: feed the hold path from BAT+ through that contact instead of from the bus. A 1 kΩ 1 W bleed resistor across the bus is the simple fix, but test it again.
Also check: K1 coil current against F4 and F7, and the DC rating of the E-stop switch contacts against that coil current. Neither is known until the parts are chosen.

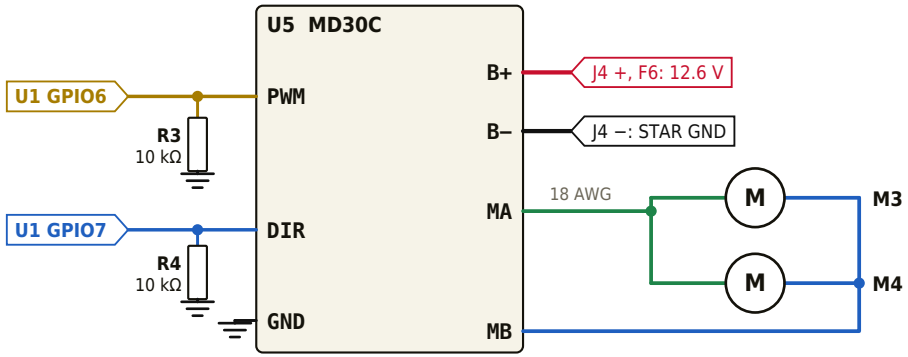
Terminal	Connects to	Terminal	Connects to
SW2 terminal 1	F4, from BAT+	K1 coil +	SW1 pole A side 2, D1 cathode
SW2 terminal 2	N3	K1 coil −	STAR GND, D1 anode
SW1 pole A, side 1	N3	K1 main contact, side A	F1 out, BAT+
SW1 pole A, side 2	K1 coil +	K1 main contact, side B	J2 +, MOTOR BUS
SW1 pole B, side 1	U1 GPIO21	SW1 pole B, side 2	Logic GND

Switch terminal marks vary. Find each pole with a meter: both must be closed in ON and open in OFF.

LEFT SIDE



RIGHT SIDE



Setup and pairing

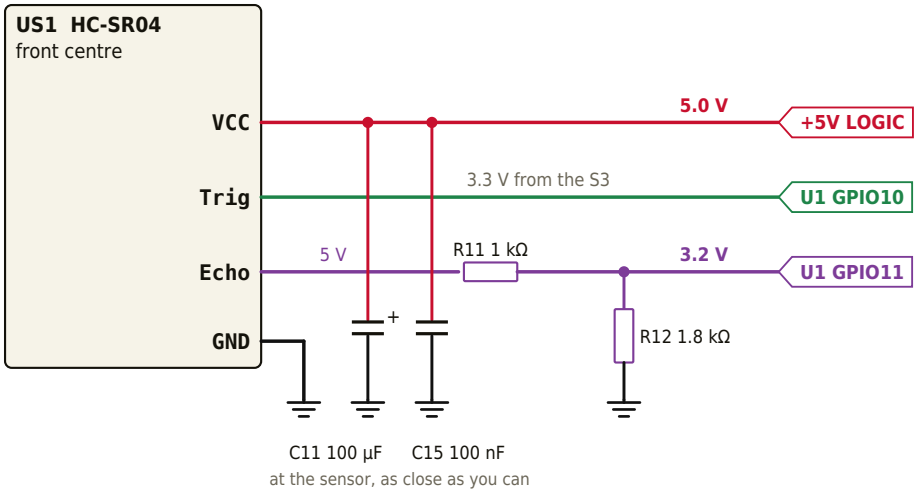
- Set the MD30C jumper or switch for external PWM (PWM + DIR, sign-magnitude). Check its manual.
- The firmware drives 20 kHz PWM at 10 bits. MD30C logic high is 3 to 5.5 V, so 3.3 V from the S3 is fine.
- The 10 kΩ pull-downs keep the driver inputs LOW while the S3 boots or resets. Solder them at the logic board.
- Check each motor alone on a bench supply first, then wire both motors on a side to spin the same way for the same polarity.
- If a whole side runs backwards, swap that side's MA and MB, or set MOTOR_LEFT_INVERT or MOTOR_RIGHT_INVERT in config_hardware.h.
- MA and MB swap polarity, so they are neither fixed + nor GND. Use a pair that is not red or black, such as green and blue, and label them.
- Route motor wires away from sonar, servo and camera-link wires (rule 3.1.5). Run motor returns to the star point, not through the logic board.

BTS7960 / IBT-2 pin (alternative driver)	Left module	Right module
RPWM	U1 GPIO4	U1 GPIO6
LPWM	U1 GPIO5	U1 GPIO7
R_EN + L_EN, both modules together	U1 GPIO15, each 10 kΩ to GND	U1 GPIO15, each 10 kΩ to GND
VCC (logic buffer)	3.3 V, see note	3.3 V, see note
B+, B−	F5, J3	F6, J4
M+, M−	M1, M2 in parallel	M3, M4 in parallel

Driver choice

To use BTS7960 modules, set MOTOR_DRIVER_TYPE to DRIVER_BTS7960 in src/s3/config_hardware.h and wire as in the table.
IBT-2 inputs go through a 74HC244 buffer. At VCC = 5 V it needs ≥3.5 V for a HIGH, which 3.3 V GPIO does not guarantee. Power the module's VCC from 3.3 V (check on your module that VCC feeds only the buffer), or add a 74HCT245.

Do not use the 10 A dual H-bridge from the team's notes with the baseline motors: one side can pull 11 A at stall.



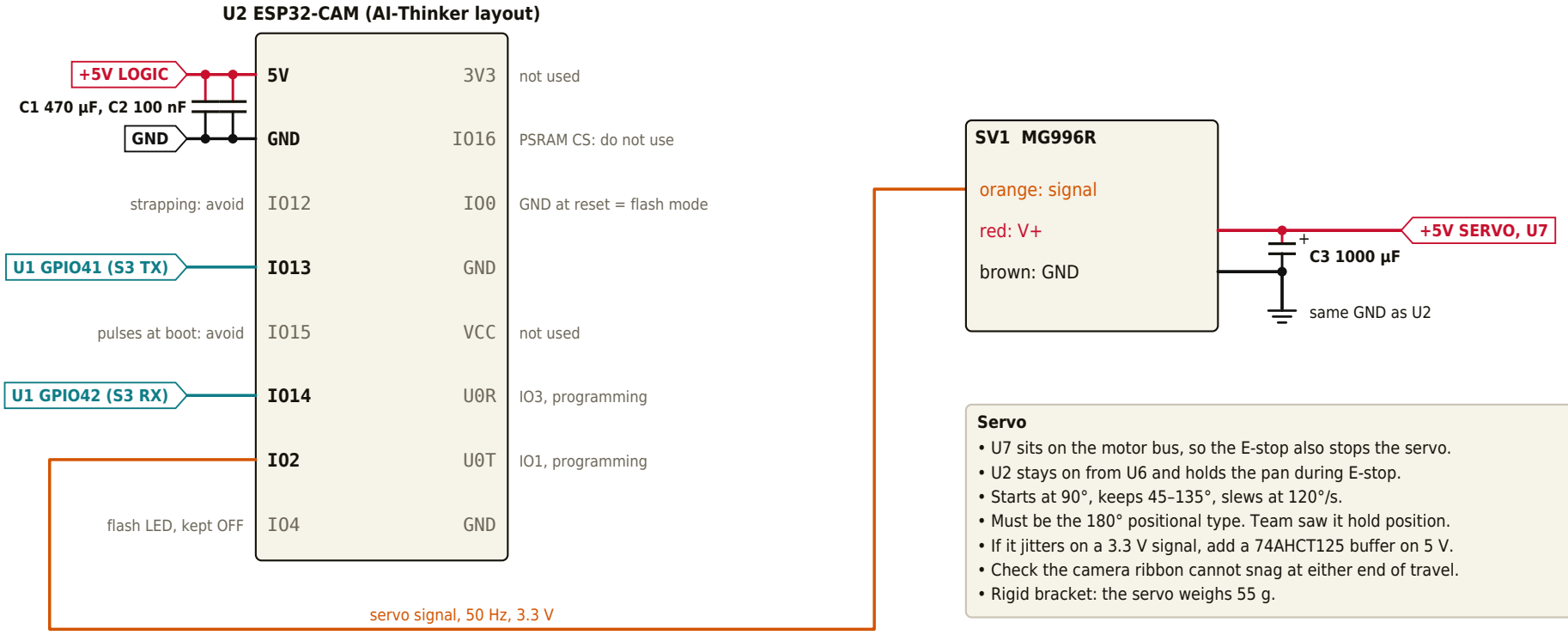
Sonar	Position	Trig	Echo	Divider	Caps
US1 front_center	forward, centred, level	GPIO10	GPIO11	R11, R12	C11, C15
US2 front_left	forward-left, about 30° out	GPIO12	GPIO13	R13, R14	C12, C16
US3 front_right	forward-right, about 30° out	GPIO14	GPIO16	R15, R16	C13, C17
US4 rear	backward. Enables reverse	GPIO17	GPIO18	R17, R18	C14, C18

Owned: US1. To buy: US2, US3, US4. Fit US1, then US4, then US2 and US3. Enable each in SONAR_FITTED.
Reverse stays blocked until the rear sonar is fitted: that is on purpose.
Sensors fire one at a time, in 40 ms slots, so they do not hear each other.
Measure each sensor face to the bumper and enter it in SONAR_FACE_TO_BUMPER_MM.

R12 may be 2 kΩ if that is what you have: 3.33 V, still safe.
NEVER connect the 5 V Echo straight to an S3 pin.

Bench result (2 Nov 2026): US1 on 3.3 V, no divider, no capacitors	Reading	Meaning
Ruler 100 mm, object propped	settled at 125 mm	25 mm long: outside the ±20 mm window
Ruler 200 mm, object propped	198 mm, then 181–191 mm	object seems to have moved in the test
Unmeasured close object, propped still	134 of 134 valid, 162–179 mm	steady at close range, no strays
Wall about 1.8 m away, nothing moving	29–45 % single stray readings	gets worse with more Wi-Fi traffic
Echo wire loose	no_echo 31 of 31	never a fake number

What to do with this
An HC-SR04 is a 5 V part, so 3.3 V is out of spec. These numbers are a warning, not a calibration. Do not calibrate on 3.3 V.
The capacitors are a proposal: Wi-Fi noise on the supply is the suspected cause of the strays, and it is not proven. Repeat the ruler test at 10, 20 and 50 cm after you build this circuit, using test T15 in Section 9, then enter corrections in SONAR_SCALE and SONAR_FACE_TO_BUMPER_MM. The proximity task must stop within 100 ± 20 mm.



Servo

- U7 sits on the motor bus, so the E-stop also stops the servo.
- U2 stays on from U6 and holds the pan during E-stop.
- Starts at 90°, keeps 45–135°, slews at 120°/s.
- Must be the 180° positional type. Team saw it hold position.
- If it jitters on a 3.3 V signal, add a 74AHCT125 buffer on 5 V.
- Check the camera ribbon cannot snag at either end of travel.
- Rigid bracket: the servo weighs 55 g.

Link wire	From	To	Colour	Signal
W60	U1 GPIO41	U2 IO13	teal (not red or black)	115200 8N1, 3.3 V both sides
W61	U2 IO14	U1 GPIO42	white (not red or black)	115200 8N1, 3.3 V both sides
W28	U1 GND	U2 GND	black	the link needs this ground

ESP32-CAM

- Leave the microSD slot EMPTY: it shares IO13 and IO14.
- The MB board is only for flashing over USB. Press IO0 + RST to flash, RST alone to run. It does not ride on the rover.
- After the first flash, updates go over Wi-Fi.
- Short, thick 5 V wires: brownouts are common on this board.

Bench result: link steady, 138 of 138 samples ok, 0 bad messages in 30 s. Servo sweep 45–135° smooth, no buzzing (team observation).

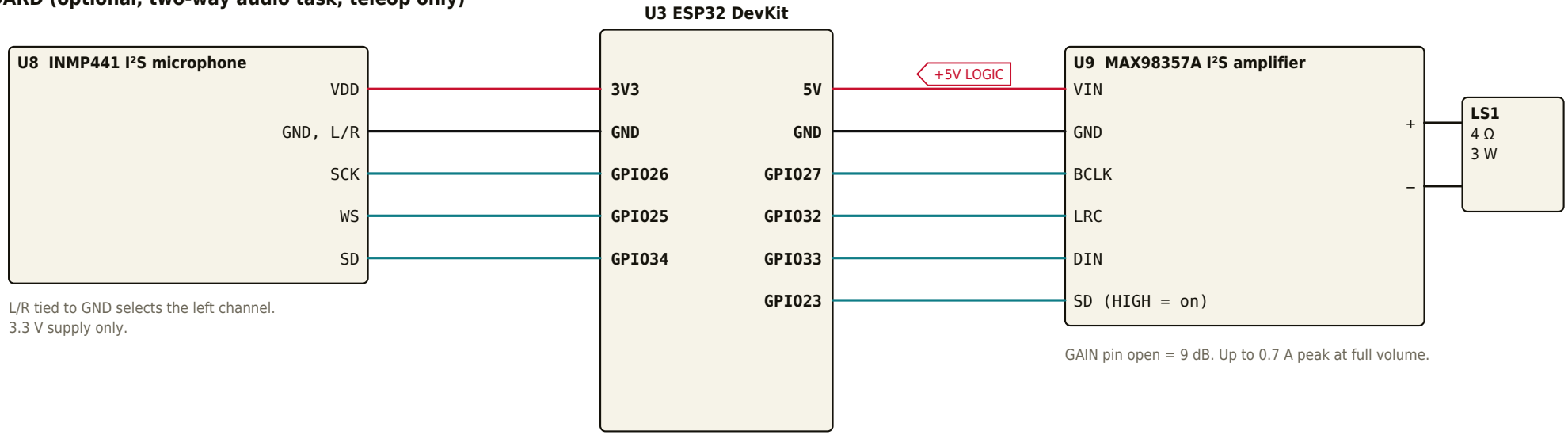
Camera was on the MB or fed from the DevKit VIN for these tests, not from U6. Repeat them once U6 is built.

VOLTAGE SENSE (optional, recommended)



12.6 V × 22 / 122 = 2.27 V at the ADC pin. Solder the divider at the logic board, then set BATTERY_SENSE_FITTED or MOTOR_BUS_SENSE_FITTED in config_hardware.h, and calibrate against a multimeter with BATTERY_CAL_GAIN. With the battery divider fitted, the firmware caps PWM duty at 12.0 V / battery voltage. Without it, duty is capped at a fixed 0.95. The bus divider lets the dashboard see that the contactor opened.

AUDIO BOARD (optional, two-way audio task, teleop only)

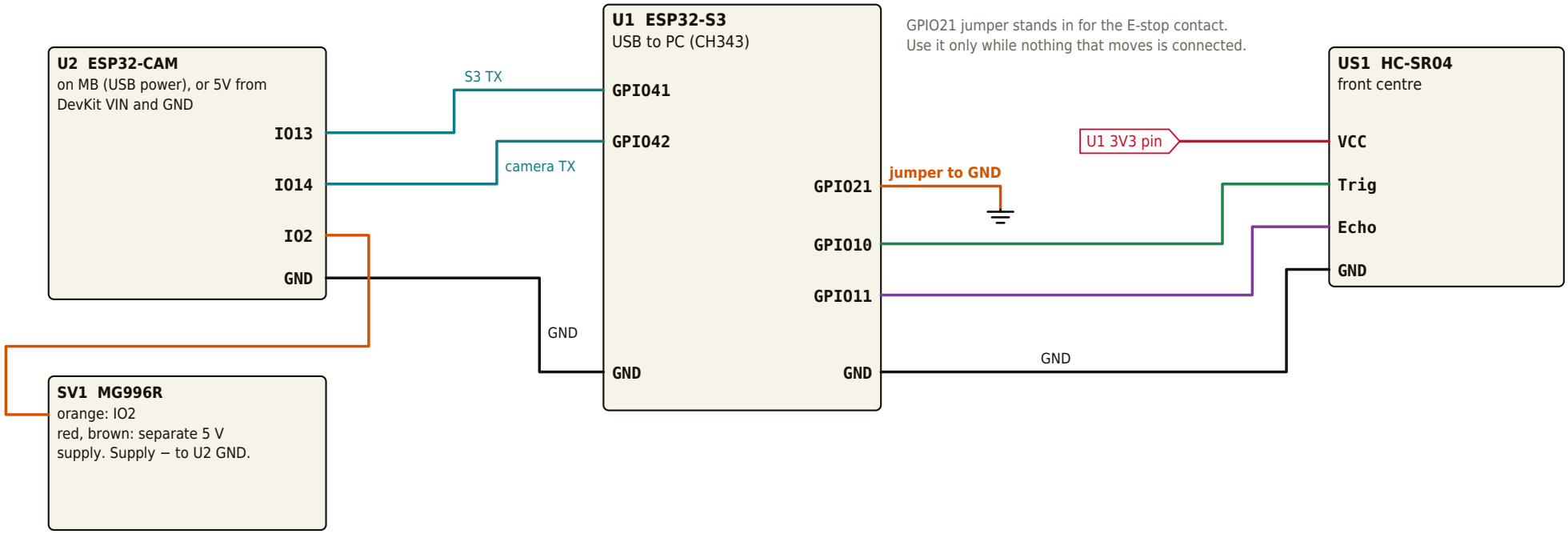


L/R tied to GND selects the left channel.
3.3 V supply only.

GAIN pin open = 9 dB. Up to 0.7 A peak at full volume.

U3 stays on its own board so audio work can never delay the motor control loop. It has no wire to the S3 except power and ground, and it talks over Wi-Fi only.
U3 and these two parts are provisional until bring-up stage 9. U3 itself runs and reports to the S3; the microphone and speaker have not been wired.

This is what is on the desk now, so you can see what has really run. Do not copy the shortcuts into the rover build.



Bench shortcut	Why it is only for the desk	On the rover
Sonar VCC on 3.3 V, Echo direct	HC-SR04 is a 5 V part; readings are 25 mm long at 100 mm	5 V + R11/R12 divider + caps (Sheet 5)
GPIO21 jumper to GND	Tells the S3 'E-stop released' with no real switch	SW1 pole B (Sheet 3)
Servo on a bench 5 V supply	No E-stop on that supply	U7 on the motor bus (Sheets 2, 6)
Camera on MB or DevKit VIN	USB or board-fed power, no capacitors	U6 + C1, C2 (Sheet 6)
Jumper wires, no solder	Rules 3.1.1 to 3.1.3 need soldered boards and crimped connectors	Veroboard or PCB, female headers

Warnings

The S3 5V pin did not power the camera on the bench, so treat it as an input only. Its use as the +5V LOGIC input is not tested. Unplug USB the first time you power it from U6.
Sonar on 5 V without a divider would feed 5 V into GPIO11 and can damage the S3. If you have no resistors yet, stay on 3.3 V and treat every reading as rough.

5. Wire schedule

Tick each wire as you make it. Servo and module pigtails keep their factory colours; extension leads follow the code.

Wire	From	To	Net and voltage	Colour	Size	Notes
W01	BT1 +	J1 + then F1 in	BAT+ 12.6 V	red	14 AWG	F1 as close to the battery as possible
W02	F1 out	K1 main contact	BAT+ 12.6 V	red	14 AWG	
W03	K1 main contact	J2 + then bus point	MOTOR BUS 0 / 12.6 V	red	14 AWG	
W04	Bus point	F5, J3 +, U4 B+	MOTOR BUS	red	14 AWG	Left driver
W05	Bus point	F6, J4 +, U5 B+	MOTOR BUS	red	14 AWG	Right driver
W06	Bus point	F8, J5 +, U7 IN+	MOTOR BUS	red	18 AWG	Servo buck
W07	F1 out	F3, J6 +, U6 IN+	BAT+ 12.6 V	red	18 AWG	Logic buck. Not switched by K1
W08	After J6 +	FAN1 +	BAT+ 12.6 V	red	22 AWG	Check fan current on its label
W09	U6 OUT+	+5V LOGIC point	+5V LOGIC	red	22 AWG	Set U6 to 5.0 V before connecting
W10	+5V LOGIC	U1 5V pin	+5V LOGIC	red	22 AWG	Unplug USB the first time
W11	+5V LOGIC	U2 5V pin, C1, C2	+5V LOGIC	red	22 AWG	Short wires
W12	+5V LOGIC	US1–US4 VCC, C11–C18	+5V LOGIC	red	24 AWG	One wire per sonar
W13	+5V LOGIC	U3 5V pin, U9 VIN	+5V LOGIC	red	22 AWG	Optional audio
W14	U7 OUT+	SV1 red, C3	+5V SERVO	red	18 AWG	Set U7 to 5.0 V before connecting
W20	BT1 −	J1 −, J2 −, star point	GND	black	14 AWG	
W21	Star point	J3 −, U4 B−	GND	black	14 AWG	Motor current return
W22	Star point	J4 −, U5 B−	GND	black	14 AWG	Motor current return
W23	Star point	J5 −, U7 IN−	GND	black	18 AWG	
W24	Star point	J6 −, U6 IN−	GND	black	18 AWG	
W25	Star point	K1 coil −, D1 anode	GND	black	22 AWG	
W26	Star point	FAN1 −	GND	black	22 AWG	
W27	Star point	Logic board GND	GND	black	22 AWG	No motor current through the logic board
W28	Logic GND	U1, U2, US1–US4 GND	GND	black	24 AWG	Also R1–R4, R12–R18, R22, R24, C21, C22
W29	Logic GND	U4 GND, U5 GND	GND	black	24 AWG	Driver logic ground
W30	U7 OUT−	SV1 brown, C3, U2 GND	GND	black	18 AWG	The servo signal needs this common ground
W31	F1 out	F4, SW2	BAT+ 12.6 V	red	22 AWG	ARM path
W32	SW2	N3	0 / 12.6 V	red	22 AWG	
W33	Bus point	F7, N3	MOTOR BUS	red	22 AWG	Hold path
W34	N3	SW1 pole A, side 1	0 / 12.6 V	red	22 AWG	
W35	SW1 pole A, side 2	K1 coil +, D1 cathode	0 / 12.6 V	red	22 AWG	
W36	U1 GPIO21	SW1 pole B, side 1	E-stop sense, 3.3 V	orange	24 AWG	Open = treated as E-stop

Wire	From	To	Net and voltage	Colour	Size	Notes
W37	SW1 pole B, side 2	Logic GND	GND	black	24 AWG	
W40	U1 GPIO4	U4 PWM	PWM 3.3 V, 20 kHz	yellow	24 AWG	R1 10 kΩ to GND
W41	U1 GPIO5	U4 DIR	DIR 3.3 V	blue	24 AWG	R2 10 kΩ to GND
W42	U1 GPIO6	U5 PWM	PWM 3.3 V, 20 kHz	yellow	24 AWG	R3 10 kΩ to GND
W43	U1 GPIO7	U5 DIR	DIR 3.3 V	blue	24 AWG	R4 10 kΩ to GND
W44	U4 MA	M1, M2 first lead	Motor, swaps polarity	green	18 AWG	Not red or black
W45	U4 MB	M1, M2 second lead	Motor, swaps polarity	blue	18 AWG	Not red or black
W46	U5 MA	M3, M4 first lead	Motor, swaps polarity	green	18 AWG	Not red or black
W47	U5 MB	M3, M4 second lead	Motor, swaps polarity	blue	18 AWG	Not red or black
W50	U1 GPIO10	US1 Trig	Trig 3.3 V	green	24 AWG	
W51	US1 Echo	R11, then U1 GPIO11	Echo 5 V, then 3.2 V	violet	24 AWG	R12 from GPIO11 side to GND
W52	U1 GPIO12	US2 Trig	Trig 3.3 V	green	24 AWG	
W53	US2 Echo	R13, then U1 GPIO13	Echo 5 V, then 3.2 V	violet	24 AWG	R14 from GPIO13 side to GND
W54	U1 GPIO14	US3 Trig	Trig 3.3 V	green	24 AWG	
W55	US3 Echo	R15, then U1 GPIO16	Echo 5 V, then 3.2 V	violet	24 AWG	R16 from GPIO16 side to GND
W56	U1 GPIO17	US4 Trig	Trig 3.3 V	green	24 AWG	
W57	US4 Echo	R17, then U1 GPIO18	Echo 5 V, then 3.2 V	violet	24 AWG	R18 from GPIO18 side to GND
W60	U1 GPIO41	U2 IO13	Link S3 to camera, 3.3 V	teal	24 AWG	115200 8N1
W61	U2 IO14	U1 GPIO42	Link camera to S3, 3.3 V	white	24 AWG	115200 8N1
W62	U2 IO2	SV1 orange	Servo signal, 50 Hz	orange	24 AWG	
W70	BAT+	R21, then U1 GPIO1	2.27 V after divider	red, then teal	26 AWG	R22, C21 to GND. Optional
W71	MOTOR BUS	R23, then U1 GPIO2	2.27 V after divider	red, then teal	26 AWG	R24, C22 to GND. Optional

6. Pin maps

U1 ESP32-S3

GPIO	Name in firmware	Direction	Connects to	Notes	Tested
1	PIN_BATT_ADC	in, ADC1	R21, R22, C21	optional	no
2	PIN_MOTOR_BUS_ADC	in, ADC1	R23, R24, C22	optional	no
4	PIN_LEFT_A	out, PWM	U4 PWM (BTS7960 RPWM)	R1 to GND	no
5	PIN_LEFT_B	out	U4 DIR (BTS7960 LPWM)	R2 to GND	no
6	PIN_RIGHT_A	out, PWM	U5 PWM (BTS7960 RPWM)	R3 to GND	no
7	PIN_RIGHT_B	out	U5 DIR (BTS7960 LPWM)	R4 to GND	no
10	PIN_SONAR_TRIG[0]	out	US1 Trig	front centre	**bench**
11	PIN_SONAR_ECHO[0]	in	US1 Echo via R11, R12	internal pull-down on	**bench**, 3.3 V
12	PIN_SONAR_TRIG[1]	out	US2 Trig	front left	no
13	PIN_SONAR_ECHO[1]	in	US2 Echo via R13, R14		no
14	PIN_SONAR_TRIG[2]	out	US3 Trig	front right	no
15	PIN_DRIVER_EN	out	BTS7960 R_EN, L_EN	BTS7960 only	no
16	PIN_SONAR_ECHO[2]	in	US3 Echo via R15, R16		no
17	PIN_SONAR_TRIG[3]	out	US4 Trig	rear	no
18	PIN_SONAR_ECHO[3]	in	US4 Echo via R17, R18		no
21	PIN_ESTOP_SENSE	in, pull-up	SW1 pole B to GND	LOW = switch ON	**bench**, jumper
41	PIN_CAM_UART_TX	out	U2 IO13	115200 8N1	**bench**
42	PIN_CAM_UART_RX	in	U2 IO14	115200 8N1	**bench**

Spare: 8, 9, 39, 40. Avoid: 0, 3, 45, 46 (strapping) · 19, 20 (USB) · 26–32 (flash) · 33–37 (octal PSRAM) · 38, 48 (RGB LED) · 43, 44 (serial monitor) · 47. ADC2 pins do not work while Wi-Fi runs, so sensing stays on ADC1.

U2 ESP32-CAM (AI-Thinker)

GPIO	Use	Connects to	Tested
13	link RX	U1 GPIO41	**bench**
14	link TX	U1 GPIO42	**bench**
2	pan servo signal, 50 Hz, strapping pin: nothing may pull it HIGH at reset	SV1 orange	**bench**
4	flash LED, kept off	none	
1, 3	programming and serial monitor	MB board only	**bench**
0	camera XCLK and boot strap: GND at reset = flash mode	camera	**bench**
32, 26, 27, 25, 23, 22, 35, 34, 39, 36, 21, 19, 18, 5	camera, on the board	camera	**bench**
16	PSRAM, do not use	none	
12, 15	avoid: 12 sets flash voltage at boot, 15 pulses at power-up	none	

U3 ESP32 DevKit (optional)

GPIO	Use	Part pin	Tested
26	I²S0 BCLK	U8 SCK	no
25	I²S0 WS	U8 WS	no
34	I²S0 data in	U8 SD	no
27	I²S1 BCLK	U9 BCLK	no
32	I²S1 LRC	U9 LRC	no
33	I²S1 data out	U9 DIN	no
23	amp enable, HIGH = on	U9 SD	no

7. Bill of materials

Ref	Part	Qty	Label	Notes
M1–M4	12 V gearmotor, Pololu #4744 (37D × 54L mm, 70:1, no encoder)	4	**RECOMMENDED** baseline. Four motors **CONFIRMED**. Team notes list 25GA 300 rpm: **UNKNOWN** rating	150 RPM, 0.2 A no-load, 5.5 A stall at 12 V. #4754 is the encoder version, needed if traverse must finish as COMPLETE.
U4, U5	Cytron MD30C, single channel, PWM + DIR	2	**RECOMMENDED**. Two drivers **CONFIRMED**, model **UNKNOWN**	5–30 V, 30 A continuous, 80 A for ≤1 s, PWM ≤20 kHz.
(alt)	BTS7960 / IBT-2	2	alternative from team notes	Firmware supports it. See Sheet 4.
BT1	3S Li-ion pack with BMS, XT60 lead	1	**RECOMMENDED**	11.1 V nominal, 12.6 V full. BMS ≥25 A continuous. For example 3S2P high-drain 18650, about 5 Ah.
—	3S Li-ion CC/CV charger, 12.6 V, 1–2 A	1	**RECOMMENDED**	Never a generic 12 V adapter.
—	Fire-resistant battery bag	1	**RECOMMENDED**	Rule 3.3.2: carry and charge in it.
SW1	2-pole ON-OFF switch (DPST), latching, labelled E-STOP, red if you can get it	1	team decision, 6 Nov 2026. **Does not meet rule 1.2.1**	Contacts DC-rated above the K1 coil current at ≥14 V DC. The single-pole KCD11 from the notes cannot both cut the coil and tell the S3.
K1	DC contactor, 12 V coil, contacts ≥40 A at ≥14 V DC	1	**RECOMMENDED**	Check the DC rating, not AC. An auxiliary NO contact is a plus (Sheet 3).
SW2	Momentary NO push button, green, ARM	1	**RECOMMENDED**	The deliberate restart (rule 1.2.4).
D1	1N4007 across K1 coil	1	**RECOMMENDED**	Cathode to coil +. Skip if K1 has a built-in diode.
F1–F8	Inline blade fuse holders and slow-blow fuses: 25 A, 15 A × 2, 3 A × 2, 1 A × 2	7	**RECOMMENDED**	ATO/ATC. F2 is not used.
J1, J2	XT60 pairs	2	**RECOMMENDED**	J2 is new compared with docs/03 first version: rule 3.2.1.
J3–J6	XT30 pairs	4	**RECOMMENDED**	Driver branches and buck inputs.
U6	5 V buck, ≥3 A continuous, 6–24 V in (logic)	1	**RECOMMENDED**	S3, ESP32-CAM, sonars, optional audio.
U7	5 V buck, ≥3 A continuous (servo)	1	**RECOMMENDED**	On the motor bus. MG996R only.
FAN1	12 V DC 4-inch case fan	1	team notes. Current **UNKNOWN**	On the F3 branch. Read its label.
U1	ESP32-S3-N16R8 board, DevKitC-1 header (marked ESP32 23 2022 V1.3)	1	**CONFIRMED** owned	Runs today on the bench.
U2	ESP32-CAM with GC2145 camera (RHYX M21-45, PID 0x2145)	1	**CONFIRMED** owned	AI-Thinker layout, confirmed by a working camera init.
—	ESP32-CAM-MB flashing board (FT232R)	1	**CONFIRMED** owned	No auto-reset: IO0 + RST to flash. Not on the rover.
US1–US4	HC-SR04 ultrasonic sensor	4	1 **CONFIRMED** owned, 3 to buy	Team notes say 2. Any subset works; rear enables reverse.
SV1	MG996R servo, 180° positional + rigid pan bracket	1	**CONFIRMED** owned	Team saw it move and hold.
R1–R4	10 kΩ pull-down at driver inputs	4	**RECOMMENDED**	Two more for R_EN and L_EN with BTS7960.
R11, 13, 15, 17	1 kΩ, Echo series	4	**RECOMMENDED**	
R12, 14, 16, 18	1.8 kΩ, Echo to GND	4	**RECOMMENDED**	2 kΩ is also fine (3.33 V).
R21, R23	100 kΩ, voltage sense top	2	**RECOMMENDED**, optional	
R22, R24	22 kΩ, voltage sense bottom	2	**RECOMMENDED**, optional	
C1	470 µF 16 V electrolytic at U2 5V	1	**RECOMMENDED**	ESP32-CAM brownouts are common.

Ref	Part	Qty	Label	Notes
C2, C15–C18, C21, C22	100 nF ceramic	7	**RECOMMENDED**	U2 5V, each sonar, each ADC pin.
C3	1000 µF 16 V electrolytic at the servo	1	**RECOMMENDED**	Servo current spikes.
C11–C14	100 µF 16 V electrolytic at each sonar	4	**RECOMMENDED**, new, not tested	Against the stray readings seen on the bench.
—	Wire: 14, 18, 22, 24 and 26 AWG	—	**RECOMMENDED**	Red and black for power, other colours for signals (rule 3.2.3).
—	Veroboard or perfboard, female headers, JST-XH or crimp connectors, heat shrink, cable ties	—	**RECOMMENDED**	Rule 3.1: no breadboard anywhere on the rover.
U3	ESP32 DevKit (D0WD-V3, 26 MHz crystal, CP2102)	1	**CONFIRMED** owned, optional	Audio only.
U8	INMP441 I²S microphone	1	team notes, provisional	Optional.
U9	MAX98357A I²S amplifier	1	team notes, provisional	Optional.
LS1	4 Ω 3 W speaker	1	team notes	Optional.

Shopping list (beyond what the team owns)

3 × HC-SR04 · rigid pan bracket · 2 × MD30C (or confirm owned drivers) · 4 × Pololu #4744 (or #4754 with encoders, or confirm the 25GA rating) · 2-pole ON-OFF E-stop switch, DC-rated above the K1 coil current · 12 V-coil contactor ≥40 A DC, ideally with an auxiliary NO contact · green NO push button · fuse holders and 25, 15, 15, 3, 3, 1, 1 A blade fuses · XT60 × 2 pairs, XT30 × 4 pairs · 2 × 5 V buck converters ≥3 A · 3S Li-ion pack with ≥25 A BMS, 12.6 V charger, fire-resistant bag · resistors: 1 k × 4, 1.8 k × 4, 10 k × 6, 100 k × 2, 22 k × 2 · capacitors: 470 µF × 1, 1000 µF × 1, 100 µF × 4, 100 nF × 7 · 1N4007 × 1 · wire · veroboard, headers, crimp connectors. Optional audio: INMP441, MAX98357A, 4 Ω 3 W speaker.

Buy first: the resistors and capacitors for the sonars. They are cheap, and the sonar work is waiting on them.

8. Calculations and build list

5 V logic budget (U6, 3 A buck)

Load	Peak at 5 V
U1 ESP32-S3, Wi-Fi peak	0.35 A
U2 ESP32-CAM	0.5 A
US1–US4	0.06 A
U3 audio board	0.3 A
U9 amplifier, full volume	0.7 A
Total	**1.9 A, leaving about 1 A margin**

Battery current, worst case

Branch	Working	At the battery
Both sides stalled	4 motors × 5.5 A	22 A
Logic, through U6	1.9 A × 5 V / 0.85 efficiency (assumed) / 11.1 V	1.0 A
Servo stalled, through U7	2.5 A × 5 V / 0.85 efficiency (assumed) / 11.1 V	1.3 A
All at once	plus fan and K1 coil, both unknown	**about 24.3 A + fan + coil**

That sits right at F1 (25 A), which is the job of F1: a full stall of everything should blow it rather than damage the wiring. Typical driving is about 2–6 A.

Dividers

Divider	Working	Result
Sonar Echo, 1 k / 1.8 k	5 V × 1.8 / (1 + 1.8)	3.21 V
Sonar Echo, 1 k / 2 k	5 V × 2 / (1 + 2)	3.33 V
Battery, 100 k / 22 k, full	12.6 V × 22 / 122	2.27 V
Battery, 100 k / 22 k, empty	9.6 V × 22 / 122	1.73 V

Motor voltage

A full 3S pack is 12.6 V, 5 % above a 12 V motor rating. The firmware caps PWM duty at 12.0 V / battery voltage once the battery divider is fitted and calibrated, and at a fixed 0.95 until then.

Build list

☐ Every module on soldered female headers on veroboard or PCB. No breadboard anywhere.
☐ Every power joint uses XT30, XT60 or Deans. No bare wire, clips or tape.
☐ Red = positive with the voltage labelled at each junction. Black = GND. Other colours for signals.
☐ Heat shrink on every joint, cable ties, strain relief at every connector.
☐ Motor wires routed away from sonar, servo and camera-link wires.
☐ E-stop rigidly mounted and reachable from outside without crossing moving parts.
☐ Battery fused at the battery, secured against vibration, in a fire-resistant bag when charging.
☐ Built circuit matches these sheets. If it does not, change the sheets before the M4 freeze.

9. Commissioning tests

Do these in order, with the wheels raised for anything that can move. Write down what you measure and send it back: nothing counts as passed until someone has seen it on the real rover. Full stages are in docs/06_bringup_guide.md.

#	Test	How	Expected	Measured	OK
T1	No short on the power rails	Battery off. Meter between BAT+ and GND, then bus and GND, then 5 V rails and GND.	No continuity beep		☐
T2	Battery voltage at F1 out	All other fuses out. Plug in the battery.	9.6–12.6 V		☐
T3	Set U6	F3 in, nothing on the U6 output. Adjust the output.	4.9–5.1 V		☐
T4	Echo divider	5 V on the sonar side of each divider. Meter on the GPIO side, board not plugged in.	3.2 V (3.33 V with 2 kΩ)		☐
T5	S3 from U6	USB unplugged. Plug in U1 only.	S3 boots, Wi-Fi joins, no restarts		☐
T6	Camera from U6	Plug in U2 with C1, C2. Watch the dashboard for 60 s.	Link ok, no restarts		☐
T7	Power on, E-stop switch ON	All fuses in, no motors, no servo. Battery on.	K1 open, bus 0 V		☐
T8	ARM	Press and release ARM.	K1 closes and stays closed, bus = battery		☐
T9	E-stop	Switch E-stop OFF.	K1 opens, bus 0 V, dashboard shows ESTOP		☐
T10	Release does not restart	Switch it back ON after a few seconds, then again about 1 s after switching OFF. Repeat after T17 with wheels raised and spinning.	K1 stays open, bus stays 0 V		☐
T11	Sense wire broken	Unplug the GPIO21 wire.	S3 latches ESTOP		☐
T12	Set U7	ARM on, servo not plugged in. Adjust the U7 output.	4.9–5.1 V		☐
T13	Servo from U7	Plug in SV1 with C3. Move the pan slider. Switch E-stop OFF while it moves.	Moves 45–135°; stops when the bus drops		☐
T14	Driver inputs low at reset	Meter on GPIO4–7 at the driver side while pressing RST on the S3. Motors off.	About 0 V		☐
T15	Sonar accuracy on 5 V	Ruler, flat object fixed in place, sonar fixed. 10, 20, 50 cm. Hands away. Close spare dashboard tabs.	Note the error. Window at 100 mm is ±20 mm		☐
T16	Voltage sense	Compare the dashboard to a meter, then set BATTERY_CAL_GAIN.	Matches the meter after calibration		☐
T17	Each motor alone	Wheels raised. One motor at a time on low duty.	Same direction per side		☐
T18	Stop test for inspection	Wheels raised, motors running. Switch E-stop OFF.	Wheels stop, bus 0 V on a meter		☐

10. What has been tested, and open problems

Run on real hardware (bench, 1–2 Nov 2026)

What	Result
S3 on USB: boot, Wi-Fi, dashboard	Runs. Telemetry 76–79 ms per frame
Camera link: GPIO41, 42 to IO13, 14	138 of 138 samples ok, 0 bad messages in 30 s
E-stop sense: GPIO21	Open: latches ESTOP at boot. Jumper to GND: released. No real switch fitted
MG996R on IO2, separate 5 V supply	Moves and holds. Sweep 45–135° smooth, no buzzing (team observation)
HC-SR04 on GPIO10, 11, on 3.3 V	Works, but 25 mm long at 100 mm, and strays at 1.8 m (Sheet 5)
AUTO QR mission	COMPLETE 4 of 4 in 2.1–4.6 s, QR held in view, camera not on the servo
Heavy web traffic	Crash found and fixed: ran 120 s under the same load, no restart

Not tested at all

Battery, BMS, fuses, XT connectors · contactor and E-stop circuit · both buck converters · S3 and camera fed from U6 · motor drivers and motors · sonars on 5 V with the divider and capacitors · sonars US2 to US4 · voltage dividers · microphone and speaker · camera mounted on the servo for the tracking task.

Open problems

Problem	Status	Next step
S3 restarted once by itself (reset reason int_wdt), idle, on earlier firmware	Cause unknown. Wiggling sonar wires did not bring it back. About 20 minutes of serial logging on the fixed firmware showed no restart	Keep the serial log on during first motor tests
Sonar reads 25 mm long at 100 mm on 3.3 V	Fails the ±20 mm window	Build Sheet 5 on 5 V, then test T15
Sonar strays at long range	Suspected supply noise, not proven	Capacitors, 5 V, then a firmware filter if still needed
Possible K1 re-latch on quick release	Design concern, not seen, not built	Test T10, then the fix on Sheet 3
S3 5V pin as a supply input	Not tested. It did not feed the camera as an output	Test T5 with USB unplugged
E-stop is a switch, not the mushroom-head button rule 1.2.1 asks for	Team decision, 6 Nov 2026	Organisers' approval in writing before inspection
Dashboard access is not protected yet	Software team	Fix before motors are connected

Revision history

Rev	Date	Change
A	2 Nov 2026	First issue. Adds J2, sonar capacitors C11–C18 and the K1 re-latch check to the design in docs/03_electrical.md.
B	6 Nov 2026	E-stop SW1 changed from the red mushroom button to a 2-pole ON-OFF switch, by team decision. Rule 1.2.1 marked NOT MET.